The Klootwijk Architectural Paradigm

Biochemical and atomic clocks
with executable self-reference

One tick. One bit toggle. One π phase advance.

$$\mathcal{T}_\pi : (N, q, \Theta) \longmapsto (N + 1,\ 1 - q,\ \Theta + \pi).$$

The clock changes a stored binary state, records that change, and reads its own record to determine the timing of the next change.

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An integrated expansion of the 22-page Version 1.0, based on the author-supplied dialogue *trewel.pdf* and the subsequent π-tick instruction.

Architectural statement

The architecture joins a structured latent capacity and an activating process. Geometry selects their encounter; chemistry and mechanics express the transition; Clock Alpha executes it; Clock Beta retains it. Each declared tick advances the encoded phase by π, complements one stored bit, and adds a record that participates in the next update.

The source names remain the organizing vocabulary: Preparaat 1, Preparaat 2, the two-sided lane, the focal cross, OTAN2, the walnut crosshatch, the hourglass, the photographic carbon copy, and the dual-clock self-reference. The new instruction specifies the clock algebra, rather than replacing these components. [S, V1, P]

What is defined, what is inherited, what is supplied

Architectural definitions assign precise meanings to the author's terms: one tick is a π increment and a Boolean complement; a stored history can become an input; a record can describe the rule that produced it.

Inherited structure is the Version 1.0 geometry, conversion, orthogonal crosshatch, Alpha/Beta dynamics, and hourglass recovery. The earlier 90° mechanical orientation is related to the new π phase by the explicit double-angle map $\theta_{\mathrm{enc}} = 2\varphi$.

Physical equations provide selectable realizations: mass-action and enzyme kinetics, transition-state rates, chemical potentials, porous transport, atomic transition phase, Rabi rotations, and Ramsey frequency discrimination. Their variables and assumptions are stated where used.

Constitutive additions specify how a particular realization connects those equations. Rate modulation by memory, lipid-order coupling, and the history-to-divider rule are declared model choices. The source provides no measured molecular species, activation barriers, material coefficients, or atomic interrogation data for a particular walnut, so the numerical instance uses explicitly selected parameters.

Operational closure

The executable instance in Sections 23–24 performs 48 deterministic, sample-and-hold ticks. Its emitted records are read back into its chemical-rate and Alpha/Beta equations. The PDF contains the specification and the resulting record; the accompanying Python program executes it. The material-field, molecular-jump, and atomic-interrogation modules are specified separately from that reduced numerical instance.

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1 The π-tick and one-bit state

Let $N \in \mathbb{N}_0$ be an event count, $q \in \{0, 1\}$ a one-bit register, and $\Theta \in \mathbb{R}$ an unwrapped architectural phase. The user's rule is the definition

$$\boxed{\mathcal{T}_\pi(N, q, \Theta) = (N + 1,\ q \oplus 1,\ \Theta + \pi).} \tag{1.1}$$

Here $\oplus$ denotes exclusive OR. Starting with $(N, q, \Theta) = (0, 0, 0)$,

$$N_n = n, \qquad q_n = n \bmod 2, \qquad \Theta_n = n\pi. \tag{1.2}$$

The circular state and a signed representation are

$$\theta_n = \Theta_n \bmod 2\pi = \pi q_n, \qquad z_n = \mathrm{e}^{\mathrm{i}\theta_n} = (-1)^{q_n}. \tag{1.3}$$

Thus a tick sends $z \mapsto -z$. Two ticks restore the bit and circular phase but add two events to the history. The full state is not restored, because N and the record have advanced.

Continuous progress, discrete events

For a nonnegative architectural angular speed $\omega_K(t)$ and $\Theta(0) = 0$, define

$$\Theta(t) = \int_0^t \omega_K(s)\,\mathrm{d}s, \qquad t_n = \inf\{t : \Theta(t) \geq n\pi\}. \tag{1.4}$$

Then $N(t) = \lfloor \Theta(t)/\pi \rfloor$ and $q(t) = N(t) \bmod 2$, using the post-tick state at a crossing. If ω_K is constant, the tick rate is $r_K = \omega_K/\pi$, while a full 2π cycle has frequency $f_K = \omega_K/(2\pi)$. Hence $r_K = 2f_K$.

Meaning of "one bit"

The register has two alternatives and a capacity of one bit. A tick performs one complement operation. This is not a claim that every predetermined toggle creates one bit of new Shannon information. With $p = \Pr(q = 1)$,

$$H(q) = -p \log_2 p - (1 - p) \log_2(1 - p) \leq 1, \qquad H(q_{n+1} \mid q_n) = 0 \tag{1.5}$$

for the deterministic rule. Random tick timing, an unknown initial state, or an external input can carry additional information in separate variables. [R14]

> **Tick identity:** one event ≡ one binary complement ≡ one π advance in the declared phase encoding. This identity fixes the architecture's counting convention.

2 Phase, mechanical angle, and quantum rotation

The original crosshatch has two designated mechanical configurations separated by 90°. [S, V1] Retain these and define the new encoded phase by

$$\varphi(q) = \frac{\pi}{2} q, \qquad \theta_{\mathrm{enc}}(q) = 2\varphi(q) = \pi q \pmod{2\pi}. \tag{2.1}$$

A mechanical quarter-turn therefore changes the encoding by a half-turn. Equation (2.1) is an explicit addition of this edition, not an equation quoted from the source.

Tick count	Bit q	Mechanical φ	Encoded θ
0	0	0	0
1	1	$\pi/2$	π
2	0	0	0
3	1	$\pi/2$	π

With $R_q = R_z(\pi q/2)$, the change of physical configuration is $R_{q_{n+1}} R_{q_n}^{\mathsf{T}}$. Successive changes are opposite quarter-turns. Repeatedly multiplying the physical frame by the same $R_z(\pi/2)$ instead produces a four-state rotation sequence; that is a different mechanism.

A literal π mechanical realization

A second admissible realization is $\varphi_\pi(q) = \pi q$. It implements a physical half-turn directly, but does not preserve the original 90° crosshatch geometry. In particular, an in-plane diagonal permeability tensor is unchanged by a 180° rotation. The main material model therefore uses Equation (2.1).

Quantum population inversion

For a quantum two-state system, a resonant rotation about the Bloch x axis is

$$R_x(\vartheta) = \exp\!\left(-\frac{\mathrm{i}\vartheta\sigma_x}{2}\right), \qquad R_x(\pi) = -\mathrm{i}\sigma_x. \tag{2.2}$$

It exchanges $|0\rangle$ and $|1\rangle$ up to a global phase. Multiplying an arbitrary state vector by $\mathrm{e}^{\mathrm{i}\pi} = -1$ alone does not exchange its populations. A classical phase inversion, a mechanical rotation, and a quantum π pulse are thus linked through specified encodings, not by identifying their different state spaces. [R8, R9]

3 Typed master state and material roles

The two preparaats are architectural roles. A realization identifies actual constituents and a selected convertible population separately. For that population,

$$P_1 = P_\star(1 - c), \qquad P_2 = P_\star c, \qquad 0 \le c \le 1. \tag{3.1}$$

In a lipid–carbohydrate realization, S denotes scaffold sites, L a mobile lipid component, and B_0, B_1 two states of an identified molecular switch. The equality $P_1 + P_2 = P_\star$ concerns the chosen switch population; it does not turn carbohydrate mass into lipid mass.

A material-and-record master state is

$$\mathsf{S} = (T, c, \ell, \mathbf{u}, p, \mathbf{C}, q, m, X, a, b, \Phi_A, N, D, \mathcal{H}, \mathcal{M}). \tag{3.2}$$

Symbol	Meaning	Unit or type
T	Temperature	K
c, ℓ, m	Switch fraction, lipid order, memory occupancy	dimensionless
$\mathbf{u}, p$	Solid displacement, pore pressure	m, Pa
$\mathbf{C}$	Chemical species concentrations	$\mathrm{mol\,m^{-3}}$
q, N	Binary state and unbounded tick count	Boolean, integer
X	Accumulated active dwell	s
a, b	Clock Alpha and Clock Beta states	normalized state
Φ_A	Atomic-reference carrier phase	rad
$D, \mathcal{H}$	Committed record and ordered history	structured data
$\mathcal{M}$	Finite description of update rules and parameters	structured data

The full state is a product of distinct spaces: adding a pressure to a bit, a phase to a concentration, or a scalar to an operator is not part of its algebra. Coupling functions must explicitly map one typed quantity into another. For example, a dimensionless occupancy may modulate a rate by $k_0/(1 + g_m m)$, where g_m is dimensionless.

Spatial and temporal phase

OTAN2's spatial bearing is written θ_{xy}; the tick phase is Θ. The atomic carrier phase is Φ_A, and the Rabi pulse area is ϑ_R. Using separate symbols makes the biochemical, geometric, and atomic connections explicit.

Inherited vocabulary and state construction: Version 1.0, Sections 1–2. New clock and model-description coordinates: this edition.

4 Two-sided lane, slit, tilt-shift, and focal cross

Use a rectangular domain $\Omega = [0, L_x] \times [0, L_y] \times [0, H]$, with opposing exchange faces Γ_0 and Γ_H. Let $I_+(z)$ enter at $z = 0$, and $I_-(z)$ enter at $z = H$. This retains the source's two-sided lane. [S, V1]

For an aperture centered at (x_0, y_0) with orientation γ, write

$$\begin{pmatrix} \xi \\ \eta \end{pmatrix} = R(-\gamma) \begin{pmatrix} x - x_0 \\ y - y_0 \end{pmatrix}, \qquad A_{w,h,\gamma}(x, y) = \mathbf{1}_{|\xi| \le w/2} \mathbf{1}_{|\eta| \le h/2}. \tag{4.1}$$

The source-named "Inuit vertical slit" is the declared narrow oriented aperture, $w \ll h$. Its action on a field E is $\mathcal{S}E = AE$. The finite aperture carries a finite width, throughput, and diffraction response; a delta function is only a specified limiting idealization.

Let $\mathcal{O}_{\phi,\mathbf{d}}$ be the calibrated tilt-shift propagation operator, with lens-plane tilt ϕ and shift $\mathbf{d}$. For a plane-to-plane projective coordinate model,

$$\widetilde{\mathbf{x}}' \sim H_{\phi,\mathbf{d}} \widetilde{\mathbf{x}}, \qquad E_{\text{out}} = \mathcal{O}_{\phi,\mathbf{d}} \mathcal{S} E_{\text{in}}. \tag{4.2}$$

The homography H is fitted or supplied with the optical realization. It is not generally the logarithmic radial map; OTAN2 supplies that separate encoding in Section 5. Geometric remapping and optical field propagation remain distinct operations. [R13]

Focal cross and the two-preparaat interaction

For equally indexed arrays of latent capacity P_1, activation P_2, and aperture/overlap mask W, the "double Hadamard intersection" is

$$C = W \odot (P_1 \odot P_2). \tag{4.3}$$

The componentwise product makes co-location explicit. Where the quantities have physical units, the rate coefficient that maps C to a reaction rate carries the corresponding units. This operator can supply an illumination input, a reaction-field source, or a digital activation mask, according to the chosen interface.

Boundary data

A complete field instance supplies $I_+(0, t)$, $I_-(H, t)$, thermal exchange conditions, scaffold constraints or tractions, and species or pressure boundary data. These are the ports through which external energy, matter, and inputs enter the architecture.

5 OTAN2 and phase-aware addressing

Retain OTAN2 as the architecture's regularized log-radius and quadrant-phase encoder, with length scales $r_0 > 0$, $\epsilon > 0$ and dimensionless radial scale $M > 0$:

$$\rho_\epsilon = \frac{M}{2} \ln \frac{x^2 + y^2 + \epsilon^2}{r_0^2}, \qquad v_\theta = \mathbf{1}_{x^2+y^2>0}. \tag{5.1}$$

For nonzero input, θ_{xy} is the quadrant-correct bearing. At $(0, 0)$, store $\theta_{xy} = 0$ with $v_\theta = 0$; finite radial encoding does not create a unique direction at the origin. [V1]

A zero-based radius and its inverse are

$$\bar{\rho} = \frac{M}{2} \ln\left(1 + \frac{x^2 + y^2}{\epsilon^2}\right), \qquad r = \epsilon\sqrt{\exp(2\bar{\rho}/M) - 1}. \tag{5.2}$$

For an octant implementation, set $a = |x|$, $b = |y|$, and, for $\max(a, b) > 0$, $z = \min(a, b)/\max(a, b)$. Let $P(z)$ approximate $\arctan z$ on $[0, 1]$ with a declared error bound. Then

$$\alpha = \begin{cases} P(z), & a \geq b, \\ \pi/2 - P(z), & b > a, \end{cases} \qquad \theta_{xy} = \begin{cases} \alpha, & x \geq 0,\ y \geq 0, \\ \pi - \alpha, & x < 0,\ y \geq 0, \\ \alpha - \pi, & x < 0,\ y < 0, \\ -\alpha, & x \geq 0,\ y < 0. \end{cases} \tag{5.3}$$

Selection masks may replace branches on a specific target. Approximation accuracy, signed-zero behavior, and execution cost are properties of that implementation, not consequences of the name OTAN2. OpenCV's documented polar/log-polar remapping is a separate reference implementation of coordinate remapping. [R13]

Direct logarithmic input

If $\ell_x = \ln|x|$ and $\ell_y = \ln|y|$ are supplied, retain the signs separately. With $\mathrm{LSE}(u_i) = m + \ln \sum_i \mathrm{e}^{u_i - m}$, $m = \max_i u_i$,

$$\rho_\epsilon = \frac{M}{2} \left[\mathrm{LSE}(2\ell_x, 2\ell_y, 2\ln\epsilon) - 2\ln r_0\right]. \tag{5.4}$$

Zero coordinates have log magnitude $-\infty$. This avoids forming a large squared Cartesian magnitude explicitly.

Clocked address

A record can index the same event as $(\bar{\rho}, \theta_{xy}, v_\theta, N, q)$. Spatial bearing and tick phase are different coordinates. A bit flip changes q and N; it changes an image-plane orientation only through the declared R_q or an explicit optical actuator.

6 Biochemical state balances for the two preparaats

Choose two identifiable switch states B_0 and B_1 with total concentration $C_\star = [B_0] + [B_1]$. The coarse-grained preparaat state is $c = [B_1]/C_\star$. Under first-order or chemostatted pseudo-first-order switching,

$$B_0 \underset{k_{10}}{\overset{k_{01}}{\rightleftharpoons}} B_1, \qquad \boxed{\dot{c} = k_{01}(1-c) - k_{10}c.} \tag{6.1}$$

Here k_{01} and k_{10} have units s^{-1}. At fixed rates,

$$c_\infty = \frac{k_{01}}{k_{01}+k_{10}}, \qquad c(t+\Delta t) = c_\infty + [c(t) - c_\infty]\mathrm{e}^{-(k_{01}+k_{10})\Delta t}. \tag{6.2}$$

The interval map is a convex combination of $c(t)$ and c_∞, so $[0,1]$ is invariant. For the convertible population, $P_1 + P_2 = P_\star$ remains constant. This is the reversible completion of the Version 1.0 conversion equation. [V1]

Thermal and photochemical channels

A selected channel decomposition is

$$k_{01} = k_{01}^{\mathrm{chem}}(T, \mathbf{a}, p) + k_{01}^{\mathrm{ph}}(T)i, \quad k_{10} = k_{10}^{\mathrm{chem}}(T, \mathbf{a}, p) + k_{10}^{\mathrm{ph}}(T)i, \tag{6.3}$$

where $\mathbf{a}$ denotes dimensionless activities and $i = (I_+ + I_-)/I_\star$ is dimensionless illumination. Alternatively, a single-molecule absorption-driven rate may be written $k^{\mathrm{ph}} = \eta\sigma_{\mathrm{abs}}\mathcal{F}$, with quantum yield η, absorption area σ_{abs}, and photon flux $\mathcal{F}$ in $\mathrm{m}^{-2}\mathrm{s}^{-1}$. The corresponding constitutive values belong to the chosen species.

General biochemical network

For stoichiometric matrix S_{ij}, concentrations C_i and elementary channel rates r_j,

$$\dot{\mathbf{C}} = \mathsf{S}\mathbf{r}(\mathbf{C}, T) + \mathbf{s}_{\mathrm{ext}}, \qquad r_j^+ = k_j^+ \prod_i C_i^{\nu_{ij}^-}, \quad r_j^- = k_j^- \prod_i C_i^{\nu_{ij}^+}. \tag{6.4}$$

The net rate is $r_j = r_j^+ - r_j^-$. Every vector $\mathbf{l}$ satisfying $\mathbf{l}^{\mathsf{T}}\mathsf{S} = 0$ identifies a conserved moiety in a closed system. The units of k_j depend on reaction order. Explicit cofactors, fuel pools, or transport terms are supplied when the selected molecular transition requires them.

> Preparaat 1 and Preparaat 2 identify paired roles. The biochemical equations additionally identify the actual molecular states, their balances, their rate laws, and their coupling to those roles.

7 Arrhenius, Eyring, and molecular timescales

The inherited temperature law is Arrhenius kinetics,

$$k(T) = A\mathrm{e}^{-E_a/(RT)}, \qquad \frac{k(T)}{k(T_r)} = \exp\left[\frac{E_a}{R}\left(\frac{1}{T_r} - \frac{1}{T}\right)\right]. \tag{7.1}$$

A first-order transition-state realization is

$$\boxed{k(T) = \kappa_{\mathrm{tr}}\frac{k_B T}{h}\exp\left(\frac{\Delta S^{\ddagger}}{R}\right)\exp\left(-\frac{\Delta H^{\ddagger}}{RT}\right) = \kappa_{\mathrm{tr}}\frac{k_B T}{h}\mathrm{e}^{-\Delta G^{\ddagger}/(RT)}.} \tag{7.2}$$

Here κ_{tr} is a transmission coefficient; activation free energies are molar. For a reaction of molecularity m written in concentrations, the corresponding standard-state expression includes $(C^{\circ})^{1-m}$. The frequency prefactor alone is not a bimolecular rate constant. [R4, R5]

Stress and memory in a selected barrier model

One declared coupling is

$$\Delta G^{\ddagger} = \Delta G_0^{\ddagger} + p\Delta V^{\ddagger} + G_c c + G_m m, \qquad \frac{\partial \ln k}{\partial p} = -\frac{\Delta V^{\ddagger}}{RT}. \tag{7.3}$$

$\Delta V^{\ddagger}$ is an activation volume in $\mathrm{m^3 mol^{-1}}$; G_c, G_m are molar energy coefficients. A positive G_m makes retained occupancy slow that channel. This is a constitutive extension, distinct from the rational rate modulation used in the executed instance.

Illustrative temperature conversion

For the selected parameters $E_a = 50\ \mathrm{kJ\,mol^{-1}}$, $T_r = 298.15\ \mathrm{K}$ and $k(T_r) = 1\ \mathrm{s^{-1}}$, Equation (7.1) gives:

T (K)	k ($\mathrm{s^{-1}}$)	Mean first-order wait $1/k$ (s)
273.15	0.157862	6.334649
288.15	0.496598	2.013702
298.15	1.000000	1.000000
308.15	1.924265	0.519679
318.15	3.553529	0.281410

These are evaluations of the selected parameters, not fitted material measurements. At 300 K, $k_B T/h = 6.2509857 \times 10^{12}\ \mathrm{s^{-1}}$; with $\Delta G^{\ddagger} = 70\ \mathrm{kJ\,mol^{-1}}$ and $\kappa_{\mathrm{tr}} = 1$, the Eyring rate is about $4.05605\ \mathrm{s^{-1}}$. The activation barrier separates a molecular frequency scale from an actual reaction rate. [R3]

8 Fuel, chemical potentials, and cycle closure

A concrete selectable biochemical module is a phosphorylatable switch. Let B_1 carry one more phosphate than B_0, and use two reversible channels:

$$B_0 + \mathrm{ATP} \rightleftharpoons B_1 + \mathrm{ADP}, \tag{8.1}$$
$$B_1 + \mathrm{H_2O} \rightleftharpoons B_0 + \mathrm{P}_i. \tag{8.2}$$

With water activity absorbed into k_b^+ and chemostatted activities,

$$r_a = k_a^+[B_0]a_{\mathrm{ATP}} - k_a^-[B_1]a_{\mathrm{ADP}}, \tag{8.3}$$
$$r_b = k_b^+[B_1] - k_b^-[B_0]a_{\mathrm{P_i}}, \tag{8.4}$$
$$[\dot{B_1}] = r_a - r_b, \quad k_{01} = k_a^+ a_{\mathrm{ATP}} + k_b^- a_{\mathrm{P_i}}, \quad k_{10} = k_a^- a_{\mathrm{ADP}} + k_b^+. \tag{8.5}$$

In a closed, well-mixed volume, $[\dot{\mathrm{ATP}}] = -r_a$, $[\dot{\mathrm{ADP}}] = r_a$, and $[\dot{\mathrm{P}}_i] = r_b$. Thus ATP plus ADP is conserved, and so is

$$3[\mathrm{ATP}] + 2[\mathrm{ADP}] + [\mathrm{P}_i] + [B_1]. \tag{8.6}$$

The coefficients count phosphate groups relative to B_0. Chemostatting instead supplies corresponding exchange fluxes.

Energetic bookkeeping

Use chemical potentials $\mu_i = \mu_i^\circ + RT \ln a_i$. The affinity of the forward cycle is

$$\mathcal{A} = \mu_{\mathrm{ATP}} + \mu_{\mathrm{H_2O}} - \mu_{\mathrm{ADP}} - \mu_{\mathrm{P_i}}. \tag{8.7}$$

Thermodynamically consistent rate constants satisfy local detailed balance. For these channels, the product of forward-to-reverse propensity ratios around the cycle is

$$\frac{k_a^+ a_{\mathrm{ATP}}\, k_b^+}{k_a^- a_{\mathrm{ADP}}\, k_b^- a_{\mathrm{P_i}}} = \mathrm{e}^{\mathcal{A}/(RT)}. \tag{8.8}$$

At a steady cycle flux J, the isothermal entropy production per volume is $J\mathcal{A}/T$. More generally, the channel expression is

$$\dot{s}_{\mathrm{prod}} = R \sum_j (r_j^+ - r_j^-) \ln(r_j^+/r_j^-) \geq 0. \tag{8.9}$$

Nonnegativity follows from $(x - y)\ln(x/y) \geq 0$ for positive x, y. This module supplies an explicit fuel balance for repeated biochemical operation; it is an added realization, not a claim that the source identified ATP-dependent walnut clock chemistry. Energy consumption and timing precision are separate measurable quantities. [R7]

9 Enzyme saturation, binding, and spatial biochemistry

For an enzyme-mediated conversion, choose

$$E + B_0 \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} EB_0 \xrightarrow{k_{\mathrm{cat}}} E + B_1. \tag{9.1}$$

Under the standard quasi-steady intermediate approximation, fixed total enzyme E_T, and the corresponding initial-rate conditions,

$$v = \frac{V_{\max}[B_0]}{K_M + [B_0]}, \qquad V_{\max} = k_{\mathrm{cat}} E_T, \qquad K_M = \frac{k_{-1} + k_{\mathrm{cat}}}{k_1}. \tag{9.2}$$

A matching reverse pathway gives $\dot{c} = (v_{01} - v_{10})/C_\star$. A first-order rate is recovered only in an appropriate dilute-substrate or pseudo-first-order limit. The source's rate symbol is thereby supplied with an explicit enzyme realization when required. [R6]

Lipid–scaffold binding

For scaffold sites S and mobile constituent L,

$$S + L \underset{k_{\mathrm{off}}}{\overset{k_{\mathrm{on}}}{\rightleftharpoons}} SL, \qquad \dot{\beta} = k_{\mathrm{on}}[L](1 - \beta) - k_{\mathrm{off}}\beta, \tag{9.3}$$

where $\beta = [SL]/S_T$. At fixed $[L]$, the equilibrium occupancy is

$$\beta_\infty = \frac{[L]}{K_d + [L]}, \qquad K_d = k_{\mathrm{off}}/k_{\mathrm{on}}. \tag{9.4}$$

Binding occupancy can modulate a rate, swelling strain, or pore permeability through declared constitutive functions. It is not the same state as the memory occupancy m unless a particular realization explicitly equates them.

Conservative porous-medium balance

Let ϕ_p be pore volume fraction, C_i the concentration per pore-fluid volume, $\mathbf{j}$ the Darcy volume flux per bulk area, and D_i a dispersion/diffusion tensor. Then

$$\begin{aligned} \partial_t(\phi_p C_i) + \nabla \cdot \mathbf{J}_i &= \sum_j \mathsf{S}_{ij} r_j + s_i, \\ \mathbf{J}_i &= C_i \mathbf{j} - \phi_p \mathsf{D}_i \nabla C_i. \end{aligned} \tag{9.5}$$

Here reaction rates are expressed per bulk volume; pore-volume rates must be multiplied by ϕ_p. The mean pore velocity is $\mathbf{j}/\phi_p$, rather than $\mathbf{j}$ itself. This balance connects chemical state to the hourglass channels while retaining mass units and boundary fluxes. [R12]

10 Lipid order, carbohydrate structure, and lever bias

Represent the source's lipid–carbohydrate pairing by a scaffold deformation $\mathbf{u}$ and a local lipid-order coordinate $\ell \in [0, 1]$. A proposed free-energy density is

$$f = W\ell^2(1-\ell)^2 + \Delta f(T, c)\ell + \frac{K_\ell}{2}|\nabla\ell|^2 + \frac{1}{2}(\varepsilon - \varepsilon^*) : \mathsf{C}_q : (\varepsilon - \varepsilon^*). \tag{10.1}$$

W and Δf are in $\mathrm{J\,m^{-3}}$, K_ℓ in $\mathrm{J\,m^{-1}}$, and $\varepsilon = \frac{1}{2}(\nabla\mathbf{u} + \nabla\mathbf{u}^{\mathsf{T}})$. The eigenstrain $\varepsilon^*(\ell, c, T)$ connects local order or activation to scaffold stress.

With $F = \int_\Omega f \, \mathrm{d}V$, a nonconserved order-relaxation law is

$$\dot{\ell} = -L_\ell \ell(1-\ell)\frac{\delta F}{\delta \ell}. \tag{10.2}$$

For fixed boundaries and other state variables,

$$\frac{\mathrm{d}F}{\mathrm{d}t} = -\int_\Omega L_\ell \ell(1-\ell)\left(\frac{\delta F}{\delta \ell}\right)^2 \mathrm{d}V \le 0. \tag{10.3}$$

If ℓ instead denotes a conserved composition, use a conservative chemical-potential flux such as $\partial_t \ell = \nabla \cdot (M_\ell \nabla(\delta F/\delta\ell))$, with its own boundary conditions. Order and composition are not silently interchanged.

Crosshatch orientation

Retain the Version 1.0 angular energy, treating an unoriented crosshatch member axis modulo π so that there are two designated wells

$$U(\varphi) = B\sin^2(2\varphi) - h_c\cos(2\varphi), \qquad B > 0. \tag{10.4}$$

The orientations $\varphi = 0, \pi/2$ are local minima when $|h_c| < 2B$: their second derivatives are $8B + 4h_c$ and $8B - 4h_c$. A selected coupling $h_c = h_0 + h_\ell \ell + h_m m$ lets chemistry and history bias the mechanical wells.

The source supplies the crosshatch-as-lever architecture, not the coefficients of this free energy. In the clocked reduced instance, the bit labels the designated configuration directly. In a resolved actuator model, $J\ddot{\varphi} + \zeta\dot{\varphi} = -\partial_\varphi U + \tau_{\mathrm{act}}$ determines whether and when a requested orientation is physically reached. [S, V1]

11 Chemical ticks: jumps, progress, and repeated cycles

There are three different, explicitly selectable ways to read a chemical process as a clock.

Molecular jump clock

For a single two-state switch, take transitions $0 \to 1$ at rate k_{01} and $1 \to 0$ at rate k_{10}. Each realized jump increments N, complements q, and advances its encoded phase by π. Constant-rate waiting times obey

$$\Pr(\tau_q > t) = \mathrm{e}^{-k_{q,1-q}t}, \qquad \mathbb{E}\tau_q = \frac{1}{k_{q,1-q}}, \qquad \mathrm{Var}(\tau_q) = \frac{1}{k_{q,1-q}^2}. \tag{11.1}$$

The stationary mean jump rate is

$$\bar{r} = \frac{2k_{01}k_{10}}{k_{01} + k_{10}}, \qquad \mathbb{E}\tau_{\mathrm{cycle}} = \frac{1}{k_{01}} + \frac{1}{k_{10}}. \tag{11.2}$$

The master equation for the probability of state 1 is exactly Equation (6.1). Individual jumps persist while the ensemble fraction relaxes to equilibrium. That ensemble equation with constant rates is not, by itself, an oscillating concentration signal.

Chemical-progress clock

For irreversible conversion, $\dot{c} = k_f(T)i(1 - c)$ gives

$$\vartheta_{\mathrm{chem}}(t) = \int_0^t k_f(T)i\,\mathrm{d}s, \qquad c(t) = 1 - [1 - c(0)]\mathrm{e}^{-\vartheta_{\mathrm{chem}}(t)}. \tag{11.3}$$

A declared transducer $\Theta = \pi\vartheta_{\mathrm{chem}}/\vartheta_\star$ yields ticks each time progress crosses another $\vartheta_\star$. A finite, one-pass substrate does not supply an indefinitely sustained macroscopic oscillator; sustained cycles require a specified reversible pathway, replenishment, or external pacing. Biochemical oscillator precision and dissipation can then be analyzed separately. [R7]

Deterministic paced clock used in this edition

The executable model uses the current outgoing rate to choose a *mean-dwell interval*, rounds that interval to an integer number of reference cycles, evolves its state, and complements its register. It does not sample the exponential waiting-time distribution. Its binary register is a paced control state, and c is its separate biochemical population coordinate. This selection makes every reported tick and replay reproducible.

> The same π-tick algebra can label random molecular jumps or a deterministic clocked controller. The event-timing law specifies which one is being executed.

12 Thermal, photonic, and depth-clock equations

For the opposed optical paths, use the inherited attenuation model

$$\mu_a(c) = \mu_1(1-c) + \mu_2 c, \quad \partial_z I_+ = -\mu_a I_+, \quad \partial_z I_- = +\mu_a I_-. \tag{12.1}$$

The sign difference follows from their opposite propagation directions. μ_1, μ_2 have units m^{-1}. The source's "Vanta-state" is the modeled large-optical-depth regime $\mathcal{T} = \int \mu_a \, \mathrm{d}z \gg 1$; its name does not assign the optical constants of a commercial coating to the material. [V1]

An enthalpy equation supplies thermal and latent-heat accounting:

$$\partial_t h_v(T, \ell) = \nabla \cdot (\kappa_T \nabla T) + \mu_a(I_+ + I_-) - \sum_j \Delta H_j r_j + Q_{\mathrm{ext}}. \tag{12.2}$$

Here h_v is enthalpy per bulk volume, κ_T thermal conductivity, ΔH_j molar reaction enthalpy, and r_j molar rate per bulk volume. A latent-heat part may be included in h_v; the same latent heat must not also be counted as an independent source.

Temperature- and depth-dependent chemical progress

For an inherited one-way exposure branch,

$$\partial_t c = k_f(T)\frac{I_+ + I_-}{I_\star}(1-c), \qquad \vartheta(z,t) = \int_0^t k_f(T(z,s))\frac{I_+(z,s) + I_-(z,s)}{I_\star}\,\mathrm{d}s. \tag{12.3}$$

The surface and core can therefore accumulate different chemical progress under the same external reference time. Clock Alpha and Clock Beta model fast and slow state response; the atomic reference supplies a separate common time coordinate.

A solvable penetration-front branch

For constant T, constant μ, one-sided $I = I_0 \mathrm{e}^{-\mu z}$, and $c(z,0) = 0$, the threshold $c(z_f, t) = c_\star$ gives

$$z_f(t) = \frac{1}{\mu}\ln\!\left(\frac{k_f I_0 t}{I_\star[-\ln(1-c_\star)]}\right), \qquad \dot{z}_f = \frac{1}{\mu t}. \tag{12.4}$$

This expression applies after the surface reaches the threshold and while $0 < z_f < H$. It makes the logarithmic penetration and inverse-time slowing precise. If absorption changes with c, the coupled equations above determine the front instead. The inverse-log *history kernel* in Section 17 is a different declared law; it is not inferred from Beer–Lambert attenuation.

13 Atomic reference: energy, phase, and the second

An atomic transition has an energy difference and associated frequency

$$\Delta E = E_e - E_g = h\nu_A = \hbar\omega_A, \qquad \omega_A = 2\pi\nu_A. \tag{13.1}$$

For the SI definition of the second, the unperturbed ground-state hyperfine transition frequency of caesium-133 is exactly

$$\boxed{\Delta\nu_{\mathrm{Cs}} = 9\,192\,631\,770\ \mathrm{Hz}.} \tag{13.2}$$

The exact defining frequency is not a measured chemical rate of the preparaats. Atomic clocks use a local oscillator interrogating an atomic transition and a feedback process that disciplines the oscillator. A counted carrier cycle is not one spontaneous atomic population jump. [R1, R2]

Reference phase and counting

For a reference oscillator with instantaneous frequency $\nu_{\mathrm{ref}}(t)$,

$$\Phi_A(t) = \Phi_A(0) + 2\pi \int_0^t \nu_{\mathrm{ref}}(s)\,\mathrm{d}s, \qquad C_A(t) = \frac{\Phi_A(t) - \Phi_A(0)}{2\pi}. \tag{13.3}$$

For the ideal fixed reference, elapsed time is $t = C_A/\Delta\nu_{\mathrm{Cs}}$. A frequency divider maps these reference cycles to the architectural phase; it need not make the chemistry run at the carrier frequency.

Exact constants and derived values

The SI constants used here are $h = 6.62607015 \times 10^{-34}$ J s, $k_B = 1.380649 \times 10^{-23}$ J K^{-1}, and $N_A = 6.02214076 \times 10^{23}$ mol^{-1}, giving $R = N_A k_B = 8.31446261815324$ J mol^{-1}K^{-1}. [R3]

From those definitions,

$$h\Delta\nu_{\mathrm{Cs}} = 6.091102297113866 \times 10^{-24}\ \mathrm{J}, \tag{13.4}$$

$$\frac{1}{2\Delta\nu_{\mathrm{Cs}}} = 5.43913878538833 \times 10^{-11}\ \mathrm{s} \approx 54.3914\ \mathrm{ps}. \tag{13.5}$$

The latter is half an ideal caesium carrier period. It is neither the duration of an atomic π pulse nor the time for a biochemical state change. The next sections specify those separate durations and their clocked connection.

14 Atomic π pulses and the physical bit flip

Choose basis states $|0\rangle$ and $|1\rangle$ of a driven atom. In a rotating-frame two-level model with the rotating-wave approximation, a selectable Hamiltonian is

$$H_R = \frac{\hbar}{2}\left[\delta\sigma_z + \Omega_R(t)(\cos\phi_d\,\sigma_x + \sin\phi_d\,\sigma_y)\right], \quad \dot{\rho} = -\frac{\mathrm{i}}{\hbar}[H_R, \rho]. \tag{14.1}$$

Ω_R is an angular Rabi frequency, δ the angular detuning, and ϕ_d the drive phase. At resonance with a fixed x-axis drive,

$$\vartheta_R(t) = \int_0^t \Omega_R(s)\,\mathrm{d}s, \qquad U_R(t) = \exp[-\mathrm{i}\vartheta_R(t)\sigma_x/2]. \tag{14.2}$$

A pulse whose area is π gives

$$\boxed{\vartheta_R(t_\pi) = \pi, \qquad U_\pi = -\mathrm{i}\sigma_x, \qquad U_\pi|q\rangle = -\mathrm{i}|1-q\rangle.} \tag{14.3}$$

This is a physical two-state realization of the architecture's complement operation. The pulse phase and population inversion follow the driven-state dynamics, not just the sign change of a global wavefunction phase. [R8, R9]

Population equation and duration

For a constant drive, starting in $|0\rangle$,

$$P_1(t) = \frac{\Omega_R^2}{\Omega_R^2 + \delta^2}\sin^2\left(\frac{\sqrt{\Omega_R^2 + \delta^2}\,t}{2}\right). \tag{14.4}$$

At resonance, $t_\pi = \pi/\Omega_R$. With the illustrative drive $\Omega_R/(2\pi) = 1$ kHz, $t_\pi = 0.5$ ms. Its duration is determined by the drive coupling, not by $1/(2\Delta\nu_{\mathrm{Cs}})$.

Pulse-completion event

An ideal driven realization declares a tick when the commanded resonant pulse completes with the required pulse area. The state change may occupy a finite interval; the event is the completion marker. A dissipative atomic instance can add a specified Lindblad operator to $\dot{\rho}$, and an actual register readout then has its corresponding population probabilities. The reduced reference code does not simulate that open-system dynamics.

> The architectural phase Θ counts completed toggles. The atomic carrier phase Φ_A supplies time. The Rabi area ϑ_R governs population rotation. They are connected, but not interchangeable.

15 Ramsey interrogation and reference-clock feedback

A two-pulse Ramsey interrogation separates population preparation from phase accumulation. In an ideal discriminator convention with contrast $C \in [0, 1]$, interrogation duration T_R, phase step ϕ, and detuning δ,

$$P_e(\phi) = \tfrac{1}{2}\left[1 + C\cos(\delta T_R + \phi)\right]. \tag{15.1}$$

The two settings $\phi = -\pi/2$ and $+\pi/2$ give an error signal

$$e = P_e(-\pi/2) - P_e(+\pi/2) = C\sin(\delta T_R). \tag{15.2}$$

Atomic-clock operation can use such a frequency discriminator to correct the local oscillator. [R10, R2]

For a declared idealized integral correction,

$$\delta_{j+1} = \delta_j - \frac{g}{T_R} e_j. \tag{15.3}$$

Linearization at the central fringe gives $\delta_{j+1} \simeq (1 - gC)\delta_j$. The local deterministic condition is $0 < gC < 2$. This condition follows from this particular one-step update; dead time, measurement noise, and actuator dynamics require their own state model.

Chemical rate and atomic frequency are separate channels

The actual interrogated transition can be represented as

$$\nu_A = \nu_0 + \Delta\nu_{\text{Zeeman}} + \Delta\nu_{\text{Stark}} + \Delta\nu_{\text{collision}} + \Delta\nu_{\text{thermal}} + \cdots. \tag{15.4}$$

This notation specifies correction channels rather than a numerical shift model. The chemical rate $k(T)$ may change exponentially over a selected temperature interval while the atomic reference is handled by its own transition and correction model. An Arrhenius law for a reaction is not automatically an Arrhenius law for the reference carrier.

Role within the architectural loop

The atomic-reference module outputs Φ_A or counted reference cycles. The biochemical module outputs an intended event rate. The divider joins them. The toggle updates q, the crosshatch reads its new configuration, and the record states which rate and divider produced the event. The reference can thereby be physically disciplined without erasing the architecture's history-dependent chemical timing.

16 Clock division and chemical-to-atomic phase mapping

For a fixed divider $d > 0$, define $\Theta = (\Phi_A - \Phi_A(0))/d$. The architectural tick rate is $2\nu_{\mathrm{ref}}/d$. More generally, each interval can use its own divider selected from the prior state.

Let $r_n > 0$ be the intended tick rate held on interval n. For a fixed reference frequency, choose an integer number of full reference cycles

$$\boxed{D_n = \left\lceil \frac{\nu_{\mathrm{ref}}}{r_n} \right\rceil, \qquad \Delta t_n = \frac{D_n}{\nu_{\mathrm{ref}}}, \qquad d_n = 2D_n.} \tag{16.1}$$

For $t \in [t_n, t_{n+1}]$, the phase connection is

$$\Theta(t) = n\pi + \frac{\Phi_A(t) - \Phi_A(t_n)}{2D_n}. \tag{16.2}$$

At t_{n+1} exactly D_n full reference cycles have elapsed, so $\Theta(t_{n+1}) = (n+1)\pi$ and the bit is complemented. Changing the divider does not create a phase discontinuity because the interval is anchored at $n\pi$.

Quantization bound

From the ceiling definition,

$$\frac{1}{r_n} \le \Delta t_n < \frac{1}{r_n} + \frac{1}{\nu_{\mathrm{ref}}}, \qquad r_n^{\mathrm{realized}} = \frac{\nu_{\mathrm{ref}}}{D_n} \le r_n. \tag{16.3}$$

The timing quantization error is smaller than one reference period, relative to the held intended interval. It does not bound chemical model error or physical-clock uncertainty. Integer cycle accumulation,

$$C_{n+1} = C_n + D_n, \qquad t_n = C_n/\nu_{\mathrm{ref}}, \qquad \Theta_n = n\pi, \tag{16.4}$$

keeps exact event and carrier-cycle counts without evaluating a sine of an enormous floating-point phase.

An optional continuously locked realization

A different, explicitly added coupling is $\dot{\Theta} = \omega_{\mathrm{chem}} + K_L \sin(\Phi_A/d - \Theta)$. With constant frequencies and $\psi = \Phi_A/d - \Theta$,

$$\dot{\psi} = \omega_A/d - \omega_{\mathrm{chem}} - K_L \sin\psi. \tag{16.5}$$

A stationary relative phase exists only when $|\omega_A/d - \omega_{\mathrm{chem}}| \le K_L$, and its branch must be selected for stability. This continuous option is not used in the executed divider model.

17 Clock Alpha, Clock Beta, and the retained core

Retain the two-scale state equations from Version 1.0:

$$\tau_\alpha \dot{a} = -a + \chi b + u, \qquad \tau_\beta \dot{b} = a - b, \qquad \tau_\beta > \tau_\alpha > 0, \quad 0 \le \chi < 1. \tag{17.1}$$

Clock Alpha is the fast active state a; Clock Beta is the slow retained state b. Their names refer to state-response timescales. Both are sampled by the same π-event counter. [S, V1]

For constant u, the equilibrium is $a_\infty = b_\infty = u/(1-\chi)$. The system matrix is

$$A = \begin{pmatrix} -1/\tau_\alpha & \chi/\tau_\alpha \\ 1/\tau_\beta & -1/\tau_\beta \end{pmatrix}. \tag{17.2}$$

Its trace is negative and its determinant $(1-\chi)/(\tau_\alpha\tau_\beta)$ positive, so both eigenvalues have negative real part. The exact held-input step is

$$\begin{pmatrix} a_{n+1} \\ b_{n+1} \end{pmatrix} = \begin{pmatrix} a_\infty \\ b_\infty \end{pmatrix} + \mathrm{e}^{A\Delta t_n} \left[\begin{pmatrix} a_n \\ b_n \end{pmatrix} - \begin{pmatrix} a_\infty \\ b_\infty \end{pmatrix} \right]. \tag{17.3}$$

The Python instance evaluates this 2×2 exponential directly.

Inverse-log readout, separate from archive storage

For prior records $D_1, \ldots, D_n$ containing scalar values c_i, define

$$w_j = \frac{1}{\ln(j + \mathrm{e})}, \qquad \bar{h}_n = \frac{\sum_{i=1}^{n} w_{n-i} c_i}{\sum_{i=1}^{n} w_{n-i}}, \qquad \bar{h}_0 = 0. \tag{17.4}$$

This normalizes the inherited inverse-log history kernel. Because $w_j > 0$, $\bar{h}_n$ lies between the smallest and largest stored c_i. The full archive $\mathcal{H}_n = (D_1, \ldots, D_n)$ is retained separately; a decaying readout weight is not a proof of permanent physical storage.

Closing the dual-clock feedback

An explicit input choice is $u_n = c_n + g_q q_n + g_h \bar{h}_n$. The history also modulates the next chemical rate in Section 23. Consequently, Clock Alpha reads both the current preparaat state and the retained record, Clock Beta integrates Alpha, and both affect later event timing through declared rate functions.

18 Crosshatch stress and clocked hourglass transport

The walnut L-system supplies a graph of crosshatch members and channels. Its embedding, material laws, and boundary conditions determine the dynamics; the production rules alone do not supply a force or a chemical rate. [S, V1]

The designated physical configuration is $R_q = R_z(\pi q/2)$, with stress and permeability transformations

$$\sigma_q = R_q \sigma_0 R_q^{\mathsf{T}}, \qquad \mathsf{K}_q = R_q \mathsf{K}_0 R_q^{\mathsf{T}}. \tag{18.1}$$

Rotation preserves stress eigenvalues, trace, and determinant. It can change tensor components relative to a fixed external pressure gradient without creating new stress eigenvalues. For $\mathsf{K}_0 = \mathrm{diag}(K_x, K_y, K_z)$, $q = 1$ exchanges the x and y permeability directions.

Material force balance and lever amplification

A selected poroelastic constitutive law is

$$\sigma = \mathsf{C}_q : [\varepsilon(\mathbf{u}) - \varepsilon^*(c, \ell, T)] - \alpha_B p \mathsf{I}, \qquad \nabla \cdot \sigma + \mathbf{f} = 0. \tag{18.2}$$

The crosshatch transmits local forces through member geometry. For an ideal lever, $F_{\mathrm{out}} l_{\mathrm{out}} = F_{\mathrm{in}} l_{\mathrm{in}}$, while compatible displacements enforce the corresponding work balance. Its centroid is

$$\mathbf{r}_{\mathrm{cm}} = \frac{\int_\Omega \rho(\mathbf{x}) \mathbf{x} \, \mathrm{d}V}{\int_\Omega \rho(\mathbf{x}) \, \mathrm{d}V}. \tag{18.3}$$

A physical rotation of that centroid requires a specified pivot and moving density distribution; the bit supplies the selected configuration, not an unexplained torque source.

Compression-dependent channel flow

Using compression magnitude $p_c = \max(0, -\operatorname{tr}\sigma/3)$, a retained constitutive choice is

$$r_p = r_{p0} \mathrm{e}^{-\beta_p p_c}, \qquad \mathsf{K} = \mathrm{e}^{-2\beta_p p_c} R_q \mathsf{K}_0 R_q^{\mathsf{T}}. \tag{18.4}$$

β_p has units Pa^{-1}. Darcy transport and fluid storage are

$$\mathbf{j} = -\frac{\mathsf{K}}{\mu_f}(\nabla p - \rho_f \mathbf{g}), \qquad S_p \dot{p} + \alpha_B \partial_t (\nabla \cdot \mathbf{u}) + \nabla \cdot \mathbf{j} = s_v. \tag{18.5}$$

μ_f is dynamic viscosity; K has units m^2; $\mathbf{j}$ has units $\mathrm{m\,s}^{-1}$. Orientation and chemical swelling now affect the same channel network through explicit interfaces. [R11]

19 Dwell, finite occupancy, and recovery

The source's "time spent X" is retained as active-state dwell,

$$X(t) = \int_0^t q(s)\,\mathrm{d}s. \tag{19.1}$$

It differs from signed fluid throughput, $M_{\mathrm{tr}}(t) = \int_0^t \int_A \rho_f \mathbf{j} \cdot \mathbf{n}\,\mathrm{d}A\,\mathrm{d}s$, and from the bounded stored fraction $m \in [0, 1]$. Those three quantities coincide only under additional reservoir and transport assumptions.

Inherited finite-memory constitutive branch

With β_t, η_t in s^{-1} and τ_r in s, retain

$$s(X) = \ln(1 + \beta_t X), \quad m(X) = 1 - \exp\left[-\frac{\eta_t X}{1 + s(X)}\right], \quad T_{\mathrm{rec}}(X) = \tau_r \ln[1 + m(X)]. \tag{19.2}$$

For $X \geq 0$, $0 \leq m < 1$ and $0 \leq T_{\mathrm{rec}} < \tau_r \ln 2$. The asymptote is approached rather than reached at a finite dwell.

A return trajectory that produces the stated recovery time is

$$\dot{m} = -\frac{1 + m}{\tau_r} \quad (m > 0), \qquad m(t) = \max\{0, (1 + m_0)\mathrm{e}^{-t/\tau_r} - 1\}. \tag{19.3}$$

Once empty, the reservoir stays at zero. The exact emptying time is $\tau_r \ln(1 + m_0)$.

Cyclic branch used by the executable instance

The reduced model uses a held-bit fill/return law,

$$\dot{m} = \begin{cases} \eta(1 - m), & q = 1, \\ -(1 + m)/\tau_r, & q = 0,\ m > 0, \\ 0, & q = 0,\ m = 0. \end{cases} \tag{19.4}$$

Its exact fill step is $m^+ = 1 - (1 - m)\mathrm{e}^{-\eta \Delta t}$; its return step is the preceding maximum expression. This cyclic branch replaces the one-pass $m(X)$ curve *for the executed instance only*. The source's dwell and recovery interpretation is retained while the interval-by-interval law is made explicit. [V1]

The memory enters the next chemical-rate denominator. Longer filling changes later tick intervals; recovery reverses that modulation. A 2π bit cycle consequently need not be a return to the same chemical or memory state.

20 Photographic commitment and model-bearing records

The inherited photographic module maps a time-dependent optical field to a committed record:

$$E_{\mathrm{exp}}(\mathbf{x}) = \int_{t_0}^{t_1} I_{\mathrm{sensor}}(\mathbf{x}, t)\, \mathrm{d}t, \qquad D = Q\mathcal{R}(E_{\mathrm{exp}}). \tag{20.1}$$

$\mathcal{R}$ is the specified sensor or material response and Q the chosen quantizer/encoder. In the architectural vocabulary, the "carbon copy", "petrification", or "flash freeze" is commitment of a time-indexed representation. The geometry and response determine what information that representation retains. [V1]

Record hold

A held-record operator satisfies $\mathrm{Freeze}(\mathrm{Freeze}(D)) = \mathrm{Freeze}(D)$. The inherited White Dove notation is the explicit record-preserving projection

$$\Psi_{\mathrm{dove}} : (D, \Delta D) \longmapsto (D, 0). \tag{20.2}$$

It is a representation operator, not a modification of biochemical rates or atomic transition energies. The executed clock appends new records; it does not activate this optional hold operator.

A record that refers to its producing model

Let $\mathcal{M}$ be the finite serialized model description, and let $h_{\mathcal{M}} = \mathrm{SHA256}(\mathrm{enc}(\mathcal{M}))$. An event payload is

$$D_{n+1}^{\mathrm{payload}} = \left(h_{\mathcal{M}}, h_n, N_{n+1}, q_{n+1}, D_n^{\mathrm{cycles}}, t_{n+1}, \mathbf{x}_{n+1}, \bar{h}_n\right). \tag{20.3}$$

Here D_n^{cycles} is the divider count, not the committed-record symbol D. The stored event adds

$$h_{n+1} = \mathrm{SHA256}(\mathrm{enc}(D_{n+1}^{\mathrm{payload}})), \qquad \mathcal{H}_{n+1} = \mathrm{append}(\mathcal{H}_n, D_{n+1}). \tag{20.4}$$

The hash excludes its own digest field; the first prior digest is $h_{\mathcal{M}}$. The model snapshot is stored once and each event points to it. Thus the record carries a finite description reference rather than containing an infinitely nested copy of itself.

The digest chain supplies a reproducible byte-level identity check for this record sequence. The model identifier covers the serialized specification; the accompanying code supplies the implementation of its named rule schema. The record is read back as numeric history in the following sections.

21 Causal self-reference under the same equations

Write the finite model description as $\mathcal{M} = (F, G, \mathcal{R}_H, C; \lambda)$: a state-flow rule, event-output rule, history readout, clock-selection rule, and parameter set. Its serialized form is a data object that can be read by the same evaluator producing the next event.

The recursive loop

For a sampled state $\mathbf{x}_n$ and prior history $\mathcal{H}_n$, define

$$\bar{h}_n = \mathcal{R}_H(\mathcal{H}_n; \mathcal{M}), \tag{21.1}$$
$$(r_n, u_n) = C(\mathbf{x}_n, q_n, \bar{h}_n; \mathcal{M}), \tag{21.2}$$
$$\mathbf{x}_{n+1} = \mathrm{Flow}_{F,\mathcal{M}}(\Delta t_n; \mathbf{x}_n, q_n, u_n, \bar{h}_n), \tag{21.3}$$
$$(N_{n+1}, q_{n+1}, \Theta_{n+1}) = (N_n + 1, 1 - q_n, \Theta_n + \pi), \tag{21.4}$$
$$D_{n+1} = G(\mathcal{M}, \mathcal{H}_n, \mathbf{x}_{n+1}, N_{n+1}, q_{n+1}), \tag{21.5}$$
$$\mathcal{H}_{n+1} = \mathrm{append}(\mathcal{H}_n, D_{n+1}). \tag{21.6}$$

The next invocation reads $\mathcal{H}_{n+1}$, which it has just produced. This is operational self-reference: prior outputs and a description of the producing model participate in the same next-update machinery. It is not only a sentence stating that the architecture is self-referential.

Causality and finite description

The readout on interval n uses records available before that interval. The record created at its end cannot change its already-completed duration. A one-step delay therefore makes the recursion causal without solving an implicit self-consistency equation.

The executable reads the model's schema name and parameters, implements the named formulas, emits its identifier with each state, and reads earlier emitted c values in Equation (17.4). It does not interpret arbitrary prose or execute arbitrary source text from the record. The mathematical rules, data representation, and executable schema are separately identifiable parts of one implementation.

Self-reference of the document

This edition specifies its own tick convention, gives the program that implements it, records that program's evaluated instance, and maps the instance back to the numbered equations. The static PDF is the committed representation. Its active realization is the evaluator together with the model snapshot and the evolving record.

The source's explicit dual-clock self-reference request occurs on page 26; the present edition supplies the causal recursive state and record mechanism.

22 Unified flow-and-tick transition

Let $\mathbf{x}$ contain the continuous states of a selected realization and let $\mathcal{H}$ contain its finite history. The full sampled state is $\mathsf{S}_n = (\mathbf{x}_n, N_n, q_n, \Theta_n, \mathcal{H}_n, \mathcal{M})$. Define a held-interval flow $\mathcal{F}_n$ and a record-and-toggle operator $\mathcal{J}_\pi$:

$$\boxed{\mathsf{S}_{n+1} = \mathcal{J}_\pi \circ \mathcal{F}_n(\mathsf{S}_n), \qquad \Delta t_n = C(\mathsf{S}_n).} \tag{22.1}$$

For a fully resolved material realization, the flow includes the following coupled blocks:

$$\partial_t h_v = \nabla \cdot (\kappa_T \nabla T) + Q_{\text{abs}} + Q_{\text{rxn}} + Q_{\text{ext}}, \tag{22.2}$$

$$\partial_t (\phi_p \mathbf{C}) = -\nabla \cdot \mathbf{J}_C + \mathsf{S}\mathbf{r} + \mathbf{s}, \tag{22.3}$$

$$\dot{c} = k_{01}(1-c) - k_{10}c, \qquad \dot{\ell} = -\mathcal{L}_\ell \frac{\delta F}{\delta \ell}, \tag{22.4}$$

$$\nabla \cdot \sigma + \mathbf{f} = 0, \qquad \mathbf{j} = -\mathsf{K}(\nabla p - \rho_f \mathbf{g})/\mu_f, \tag{22.5}$$

$$\dot{X} = q, \qquad \dot{m} = f_m(q, m, \mathbf{j}), \tag{22.6}$$

$$\dot{a} = (-a + \chi b + u)/\tau_\alpha, \qquad \dot{b} = (a - b)/\tau_\beta, \tag{22.7}$$

$$\dot{\Phi}_A = 2\pi \nu_{\text{ref}}, \qquad \dot{\Theta} = \pi/\Delta t_n. \tag{22.8}$$

Here $\mathcal{L}_\ell = L_\ell \ell(1-\ell)$. When c is already a component ratio of $\mathbf{C}$, its displayed equation is that ratio's induced balance, not an extra independent copy of the reaction.

Event completion

At the interval endpoint, $\mathcal{J}_\pi$ increments N, complements q, selects the associated mechanical configuration, and appends the output record. Continuous chemical states are carried through unless a particular impulsive actuator or reaction law explicitly specifies a jump. The clock event alone does not instantaneously convert an entire macroscopic population.

Closure requirements

Every realized block specifies its initial state, boundary ports, coefficients, and map to the next block. The physical continuum model and the reduced executable are two different selections of this common typed architecture. The following instance fixes all parameters required for its lumped chemical, memory, dual-clock, divider, and record states. It does not substitute selected numerical values for unprovided optical or material-field measurements.

For fixed q the flow evolves the state; at a tick the bit changes; after commitment the updated state becomes input. This separates continuous evolution, binary event algebra, and history recursion while keeping them in one composition.

23 Fully specified self-referential clock instance

The reference program selects the reduced state $\mathbf{x} = (c, a, b, m, X)$ and uses a deterministic held-rate schedule. All states, rates, and the history readout used on an interval are evaluated from its start, unless they appear in the explicit exact-flow update.

Set $T = T_r = 298.15$ K, $E_a = 50\,000$ J mol^{-1} and $k_r = 1$ s^{-1}, with

$$k_T = k_r \exp\left[\frac{E_a}{R}\left(\frac{1}{T_r} - \frac{1}{T}\right)\right], \tag{23.1}$$

$$k_{01,n} = k_T \frac{1 + 0.35a_n + 0.25\bar{h}_n}{1 + 0.5m_n}, \tag{23.2}$$

$$k_{10,n} = 0.6k_T \frac{1 + 0.2b_n}{1 + 0.5m_n}, \tag{23.3}$$

$$r_n = (1 - q_n)k_{01,n} + q_n k_{10,n}, \qquad u_{\mathrm{ref}} = 9\,192\,631\,770 \text{ Hz}. \tag{23.4}$$

D_n and Δt_n are obtained from Equation (16.1). The two-state fraction follows Equation (6.2) with these held rates. The binary register is not sampled from this fraction; it is the paced control state.

The Alpha/Beta parameters and held input are

$$\tau_\alpha = 0.2 \text{ s}, \quad \tau_\beta = 4 \text{ s}, \quad \chi = 0.2, \qquad u_n = c_n + 0.2q_n + 0.2\bar{h}_n. \tag{23.5}$$

Their exact step is the matrix-exponential map of Section 17. The memory parameters are $\eta = 0.45$ s^{-1}, $\tau_r = 5$ s, with

$$m_{n+1} = \begin{cases} 1 - (1 - m_n)\mathrm{e}^{-0.45\Delta t_n}, & q_n = 1, \\ \max\{0, (1 + m_n)\mathrm{e}^{-\Delta t_n/5} - 1\}, & q_n = 0. \end{cases} \tag{23.6}$$

Dwell accumulates as $X_{n+1} = X_n + q_n\Delta t_n$. Initialization is

$$(c_0, a_0, b_0, m_0, X_0) = (0.1, 0, 0, 0, 0), \qquad N_0 = q_0 = C_0 = 0, \qquad \mathcal{H}_0 = (). \tag{23.7}$$

After every interval, the register flips and a state record is appended. The history readout in the next interval includes that newly emitted record.

Where self-reference changes the calculation

The coefficient $0.25\bar{h}_n$ changes the forward rate, and $0.2\bar{h}_n$ changes the Alpha input. Both use the model's own previous output concentrations. The read model snapshot fixes the rule schema and numerical parameters; each event identifies that snapshot. This instance therefore combines self-description with causal output feedback, rather than using a record solely for display.

24 Executed values and self-reference comparison

The accompanying program was executed for 48 ticks using the parameter file reproduced in Appendix B. It was then executed a second time from the same initial data; the canonical record sequences matched. These are calculated trajectories of the stated deterministic instance.

n	t_n (s)	q_n	c_n	a_n	b_n	m_n
1	1.000000	1	0.519004	0.102147	0.018298	0.000000
2	2.660589	0	0.651881	0.871380	0.274759	0.526340
3	3.529588	1	0.685990	0.843800	0.385787	0.282836
4	5.295684	0	0.691891	1.134179	0.636552	0.676059
8	10.513252	0	0.682249	1.223613	1.005047	0.752355
16	20.737014	0	0.675761	1.251441	1.172195	0.755818
24	30.865817	0	0.675034	1.254821	1.191947	0.754263
32	40.980483	0	0.674964	1.255363	1.194468	0.754042
40	51.093043	0	0.674968	1.255522	1.194884	0.754026
48	61.205111	0	0.674976	1.255607	1.195013	0.754033

The first interval is exactly 1 s. The next is approximately 1.660589 s because the outgoing reverse rate differs from the initial forward rate. Later timing depends on chemical population, Alpha/Beta state, occupancy, and the inverse-log readout of earlier records.

Removing only the explicit history feedback

A comparison run sets the two history coefficients to zero: $g_{\text{history}} = 0$ in k_{01} and $g_{\text{input,history}} = 0$ in u. The archive and all other equations remain present. The calculated 48-tick completion times are

$$t_{48}^{\text{history on}} = 61.2051105903 \text{ s}, \tag{24.1}$$

$$t_{48}^{\text{history off}} = 64.6800511809 \text{ s}, \tag{24.2}$$

$$t_{48}^{\text{on}} - t_{48}^{\text{off}} = -3.4749405906 \text{ s}. \tag{24.3}$$

This difference is an executed consequence of the selected feedback coefficients. Its sign and magnitude are not universal properties of every biochemical or atomic realization.

At tick 48, $q = 0$ and the encoded circular phase has returned to zero after 24 full cycles. The unwrapped phase is 48π, accumulated active dwell is approximately 40.472793 s, and $m \approx 0.754033$. The record and continuous state have advanced even though the bit has returned to its initial value.

Executed closure: the emitted state changes the next rate; the rate changes the integer divider; the divider fixes the next π endpoint; the complemented bit changes the next fill or recovery branch; the new state is recorded and read again.

25 Derived invariants and executed checks

1. Tick parity and two-cycle structure

Induction on Equation (1.1) gives $q_n = q_0 \oplus (n \bmod 2)$ and $\Theta_n = \Theta_0 + n\pi$. A pair of toggles is the identity on the bit, but adds two to N. The history append makes the full map history-bearing, not a two-state identity.

2. Phase endpoint and reference-cycle identity

On interval n, $\Delta\Phi_A = 2\pi D_n$. Dividing by $2D_n$ yields exactly $\Delta\Theta = \pi$. Summing the integer dividers gives the integer carrier-cycle account $C_n = \sum_{j<n} D_j$.

3. Population, memory, and readout bounds

Equation (6.2) preserves $c \in [0, 1]$. Both exact memory branches preserve $m \in [0, 1]$. Positive normalized history weights imply $\bar{h}_n \in [0, 1]$ when the stored fractions are in that interval. With nonnegative initial Alpha/Beta states, nonnegative held input and $0 \le \chi < 1$, the cooperative linear flow preserves their nonnegativity.

4. Stable held-input dual-clock flow

The trace/determinant argument in Section 17 proves asymptotic stability for a fixed held input. The proof concerns that linear flow, not an assertion of global stability for every nonlinear history-feedback coefficient set.

5. Record reproducibility

Every event digest was recomputed from its canonical payload, its predecessor digest was checked, and its model identifier matched the serialized input. Two independent calls to the reference simulation produced identical canonical records in this execution environment.

Executed test	Result
48 tick indices, parity and phase counts	Pass
Integer divider / accumulated carrier-cycle accounting	Pass
All c and m values in $[0, 1]$	Pass
Maximum computed $\lvert(1 - c) + c - 1\rvert$	0.0
Model identifier and predecessor digest chain	Pass
Canonical deterministic replay	Match

The zero population residual is the result of this arithmetic check; it is not a measurement of biochemical conservation error. Complete event records and machine-readable checks accompany the PDF.

26 Source concordance and edition map

This edition is grounded in the source dialogue and its Version 1.0 formalization. The following map separates inherited content from the newly specified connections. Source page numbers refer to the 49-page *trewel.pdf*, not to this document.

Source element	Location	Formal role in this edition
OTAN2; OpenCV/DSP	pp. 3–6, 13–14	Regularized radius, quadrant bearing and validity; Section 5.
Preparaat 1 and 2; opposed exposure	pp. 6–12	Typed roles and conversion/absorption equations; Sections 3, 6, 12.
Slit and tilt-shift	pp. 14–16	Finite aperture and declared optical map; Section 4.
Walnut crosshatch; 90° configuration	pp. 16–19	Graph, lever, rotated configuration; Sections 2, 10, 18.
Hourglass channels and recovery	pp. 18–23	Stress-coupled flow, dwell and occupancy; Sections 18–19.
Temperature and chemical clock	pp. 23–25	Arrhenius progress; new molecular and fuel modules in Sections 7–11.
Clock Alpha/Beta self-reference	pp. 26–29	Dual-scale flow, normalized history, causal record feedback; Sections 17, 21.
“Ticking means 1bit means flipping over pi”	Current instruction	New tick axiom and explicit double-angle bridge; Sections 1–2.

New realization choices

The ATP switch, enzyme binding, lipid-order free energy, atomic Rabi/Ramsey realization, divider law, self-describing event format, and numerical coefficients are supplied here as explicit extensions. They are not quotations of unprovided molecular measurements. The reference values are supported by the cited primary and official sources; the cross-domain couplings are the equations declared in this edition.

What the executable evaluates

The code evaluates the reduced two-state population, Alpha/Beta flow, cyclic memory, reference-cycle divider, tick parity, and history record. It does not evaluate the continuum optical, thermal, mechanical, molecular-noise, or atomic-interrogation modules. Those modules are specified to make their interfaces available for a material realization without relabeling a lumped simulation as an experiment.

Final architectural identity

$$\Delta\Theta = \pi \quad \Longleftrightarrow \quad q \mapsto q \oplus 1, \qquad \mathcal{H}_{n+1} = \mathcal{H}_n \| D_{n+1}.$$

The bit flips, the count advances, and the next update reads the history of that same operation.

References and source basis

[S] Tom Klootwijk, author-supplied dialogue, *trewel.pdf*, 49 pages. Source prompts and the accompanying dialogue are distinguished by the page concordance in Section 26.

[V1] *The Klootwijk Architectural Paradigm*, attributed to Tom Klootwijk, NL200678942 / 10-07-1990, Version 1.0, 16 October 2026, 22 pages. Preceding formalization supplied in this conversation.

[P] Tom Klootwijk, subsequent instruction in this conversation: add biochemical and atomic-clock equations and make the architecture self-referential, with a tick interpreted as a one-bit flip over π.

[R1] Bureau International des Poids et Mesures, *SI base unit: second*. Official definition fixing the caesium-133 hyperfine transition frequency. Accessed 16 October 2026.

[R2] National Institute of Standards and Technology, *How Do Atomic Clocks Work?* Official account of the local oscillator, atomic interrogation, and feedback. Accessed 16 October 2026.

[R3] National Institute of Standards and Technology, *Meet the Constants*. Exact defining constants of the SI, including h, k_B, and N_A. Accessed 16 October 2026.

[R4] H. Eyring, "The Activated Complex in Chemical Reactions," *Journal of Chemical Physics* **3**, 107–115 (1935). DOI: 10.1063/1.1749604.

[R5] L. Bonnet, *Eyring equation and the second order rate law: Overcoming a paradox*, arXiv:1603.01662v1 (2016). Primary derivation addressing molecularity and transition-state rate laws.

[R6] IUB/IUPAC, *Symbolism and Terminology in Enzyme Kinetics: Recommendations 1981*, Sections 4–6. Official enzyme-kinetics definitions, rate equations, and Michaelis–Menten parameters.

[R7] R. Marsland III, W. Cui and J. M. Horowitz, *The Thermodynamic Uncertainty Relation in Biochemical Oscillations*, arXiv:1901.00548. Primary treatment of biochemical clock fluctuations, precision, and thermodynamic cost.

[R8] Massachusetts Institute of Technology OpenCourseWare, *5.61 Physical Chemistry, Lecture 36: Time Dependence of Two-Level Systems: Density Matrix, Rotating Wave Approximation*, Fall 2017. Driven two-state dynamics and π-pulse population inversion; see especially lecture pages 6–8.

[R9] B. Zwiebach, Massachusetts Institute of Technology OpenCourseWare, *8.05 Quantum Physics II, Chapter 7: Two-state systems*, Fall 2013. Two-level dynamics, Pauli operators, and rotations.

[R10] M. Shuker et al., *Ramsey Spectroscopy with Displaced Frequency Jumps*, NIST publication record (2019). Primary work on Ramsey frequency discrimination and clock control. The ideal discriminator and one-step servo used here are stated in Section 15.

[R11] COMSOL, *Darcy's Law*, version 6.3 documentation. Primary technical definitions for pressure-driven porous flow and permeability.

[R12] COMSOL, *Transport in Porous Media*, version 6.3 documentation. Conservative species balance and Darcy versus interstitial velocity conventions.

[R13] OpenCV, *Geometric Image Transformations*, documentation retrieved as version 4.13.0. Polar/log-polar remapping and projective transforms; not a source for the architecture-specific name OTAN2.

[R14] C. E. Shannon, *A Mathematical Theory of Communication*, *Bell System Technical Journal* **27**, 379–423 and 623–656 (1948). Primary source for entropy and binary information units.

A Executable reference implementation

The complete program below uses Python 3.9 or later and only the standard library. It reads the model snapshot, performs the exact held-interval updates, verifies the emitted records, reruns for reproducibility, and evaluates the zero-history-gain comparison. Run it from the reference package directory:

```
python pi_clock.py --model model.json --out results
```

The outputs are `events.csv`, `events.json`, `checks.json`, and `model_snapshot.json`. The output's `theta_over_pi` field stores the integer N rather than a rounded floating-point multiple of π. The `atomic_cycles` field stores the accumulated integer carrier-cycle count.

Each interval uses the prior state for its held rates and input, then writes the updated state. The state bounds in the code express the mathematical domains of concentration and occupancy.

```python
#!/usr/bin/env python3
"""Klootwijk pi-clock: deterministic, sample-and-hold reference model.
Run: python pi_clock.py --model model.json --out results
Python 3.9+, standard library only. Equation definitions are in the PDF.
"""
from __future__ import annotations

import argparse
import csv
import hashlib
import json
import math
from pathlib import Path
from typing import Any

R_GAS = 8.31446261815324
SCHEMA = "sample-hold-pi-toggle-v1"


def canonical(value: Any) -> bytes:
    return json.dumps(value, sort_keys=True, separators=(",", ":"),
                      ensure_ascii=True, allow_nan=False).encode("utf-8")


def digest(value: Any) -> str:
    return hashlib.sha256(canonical(value)).hexdigest()


def read_model(path: Path) -> dict:
    model = json.loads(path.read_text(encoding="utf-8"))
    if model.get("rule_schema") != SCHEMA:
        raise ValueError("Unsupported rule_schema")
    p = model["parameters"]
    if not all(math.isfinite(float(v)) for v in p.values()):
        raise ValueError("Parameters must be finite")
    positive = ("nu_ref", "temperature", "temperature_ref", "k_ref",
                "reverse_ratio", "tau_alpha", "tau_beta", "tau_recovery")
    if any(p[k] <= 0 for k in positive):
        raise ValueError("Positive reference, temperature and rates required")
    if not 0 <= p["chi"] < 1:
        raise ValueError("chi must be in [0, 1)")
    nonnegative = ("activation_energy", "g_alpha", "g_beta", "g_history",
                   "g_memory", "input_q", "input_history", "fill_rate")
    if any(p[k] < 0 for k in nonnegative):
        raise ValueError("Selected model requires nonnegative coefficients")
    s = model["initial"]
    if any(not math.isfinite(float(v)) for v in s.values()):
        raise ValueError("Initial states must be finite")
    if not 0 <= s["c"] <= 1 or not 0 <= s["m"] <= 1:
        raise ValueError("Initial c and m must be in [0, 1]")
    if min(s["a"], s["b"], s["dwell"]) < 0:
        raise ValueError("Initial a, b and dwell must be nonnegative")
    if not isinstance(model["ticks"], int) or model["ticks"] < 1:
        raise ValueError("ticks must be a positive integer")
```

```python
    return model


def history_readout(records: list[dict]) -> float:
    if not records:
        return 0.0
    weights = [1.0 / math.log(j + math.e)
               for j in range(len(records))]
    values = [row["c"] for row in reversed(records)]
    return math.fsum(w*v for w, v in zip(weights, values)) / math.fsum(weights)


def dual_clock(a: float, b: float, u: float,
               dt: float, p: dict) -> tuple[float, float]:
    """Exact 2x2 matrix exponential for constant held input."""
    aa, ab = -1/p["tau_alpha"], p["chi"]/p["tau_alpha"]
    ba, bb = 1/p["tau_beta"], -1/p["tau_beta"]
    half_trace = (aa + bb)/2
    spread = math.sqrt(((aa-bb)/2)**2 + ab*ba)
    ep = math.exp((half_trace + spread)*dt)
    em = math.exp((half_trace - spread)*dt)
    c0 = (ep + em)/2
    c1 = ((ep-em)/(2*spread) if spread else ep*dt)
    eq = u/(1-p["chi"])
    da, db = a-eq, b-eq
    an = eq + c0*da + c1*((aa-half_trace)*da + ab*db)
    bn = eq + c0*db + c1*(ba*da + (bb-half_trace)*db)
    return an, bn


def simulate(model: dict) -> tuple[list[dict], dict]:
    p, state = model["parameters"], dict(model["initial"])
    model_id = digest(model)
    previous = model_id
    cycles, q = 0, 0
    records: list[dict] = []
    temperature_factor = math.exp(p["activation_energy"]/R_GAS *
                          (1/p["temperature_ref"] - 1/p["temperature"]))
    kt = p["k_ref"]*temperature_factor
    for n in range(model["ticks"]):
        hbar = history_readout(records)
        c, a, b, m = (state[k] for k in ("c", "a", "b", "m"))
        denominator = 1 + p["g_memory"]*m
        k01 = kt*(1 + p["g_alpha"]*a + p["g_history"]*hbar)/denominator
        k10 = kt*p["reverse_ratio"]*(1 + p["g_beta"]*b)/denominator
        rate = k01 if q == 0 else k10
        divider = math.ceil(p["nu_ref"]/rate)
        dt = divider/p["nu_ref"]
        total = k01+k10
        ceq = k01/total
        cn = ceq + (c-ceq)*math.exp(-total*dt)
        u = c + p["input_q"]*q + p["input_history"]*hbar
        an, bn = dual_clock(a, b, u, dt, p)
        if q == 1:
            mn = 1 - (1-m)*math.exp(-p["fill_rate"]*dt)
        else:
            mn = max(0.0, (1+m)*math.exp(-dt/p["tau_recovery"])-1)
        dwell = state["dwell"] + q*dt
        state = dict(c=cn, a=an, b=bn, m=mn, dwell=dwell)
        cycles += divider
        row = dict(n=n+1, q_before=q, q=1-q, theta_over_pi=n+1,
                   atomic_cycles=cycles, divider_cycles=divider,
                   t=cycles/p["nu_ref"], dt=dt, rate=rate,
                   k01=k01, k10=k10, hbar=hbar, held_input=u,
                   **state, model_id=model_id, previous_digest=previous)
        row["digest"] = digest(row)
        records.append(row)
        previous, q = row["digest"], 1-q
    checks = verify(model, records)
    return records, checks


def verify(model: dict, rows: list[dict]) -> dict:
    previous = digest(model)
```

```
    cycles = 0
    residual = 0.0
    for n, row in enumerate(rows, 1):
        payload = {k:v for k,v in row.items() if k != "digest"}
        assert digest(payload) == row["digest"]
        assert row["previous_digest"] == previous
        assert row["model_id"] == digest(model)
        assert row["n"] == row["theta_over_pi"] == n
        assert row["q"] == (n % 2) == 1-row["q_before"]
        assert row["divider_cycles"] >= 1
        cycles += row["divider_cycles"]
        assert cycles == row["atomic_cycles"]
        assert 0 <= row["c"] <= 1 and 0 <= row["m"] <= 1
        residual = max(residual, abs((1-row["c"])+row["c"]-1))
        previous = row["digest"]
    return dict(ticks=len(rows), parity=True, phase_count=True,
                atomic_cycle_accounting=True, record_chain=True,
                bounded_populations=True, max_population_residual=residual,
                final_digest=previous, final_time_s=rows[-1]["t"])


def main() -> None:
    parser = argparse.ArgumentParser(description=__doc__)
    parser.add_argument("--model", type=Path, default=Path("model.json"))
    parser.add_argument("--out", type=Path, default=Path("results"))
    args = parser.parse_args()
    model = read_model(args.model)
    rows, checks = simulate(model)
    repeated, _ = simulate(model)
    checks["deterministic_replay"] = (canonical(rows) == canonical(repeated))
    control = json.loads(json.dumps(model))
    control["parameters"]["g_history"] = 0.0
    control["parameters"]["input_history"] = 0.0
    control_rows, _ = simulate(control)
    checks["history_off_final_time_s"] = control_rows[-1]["t"]
    checks["history_feedback_time_difference_s"] = (
        rows[-1]["t"] - control_rows[-1]["t"])
    assert checks["deterministic_replay"]
    args.out.mkdir(parents=True, exist_ok=True)
    with (args.out/"events.csv").open("w", newline="", encoding="utf-8") as f:
        writer = csv.DictWriter(f, fieldnames=list(rows[0]))
        writer.writeheader()
        writer.writerows(rows)
    for name, obj in (("events.json", rows), ("checks.json", checks),
                      ("model_snapshot.json", model)):
        (args.out/name).write_text(json.dumps(obj, indent=2)+"\n", encoding="utf-8")
    print(json.dumps(checks, indent=2))


if __name__ == "__main__":
    main()
```

B Model snapshot and reproducible record

The following finite description is read by the executable. The `rule_schema` selects the implemented update convention; the `rules` entries describe its equations; the `parameters` and `initial` entries supply its evaluated data. The schema is an implemented contract rather than an interpreter of arbitrary text.


```
{
  "architecture": "The Klootwijk Architectural Paradigm",
  "edition": "2.0-pi-clock",
  "author": "Tom Klootwijk",
  "rule_schema": "sample-hold-pi-toggle-v1",
  "rules": {
    "tick": "N_next=N+1; q_next=1-q; Theta_next=pi*N_next",
    "reference": "D=ceil(nu_ref/r); dt=D/nu_ref",
    "chemistry": "dc/dt=k01*(1-c)-k10*c",
    "alpha": "tau_alpha*da/dt=-a+chi*b+u",
    "beta": "tau_beta*db/dt=a-b",
    "readout": "hbar=sum(c_i/log(age_i+e))/sum(1/log(age_i+e))",
    "record": "append model_id, previous_digest, tick and updated state"
  },
  "parameters": {
    "nu_ref": 9192631770,
    "temperature": 298.15,
    "temperature_ref": 298.15,
    "activation_energy": 50000.0,
    "k_ref": 1.0,
    "reverse_ratio": 0.6,
    "tau_alpha": 0.2,
    "tau_beta": 4.0,
    "chi": 0.2,
    "g_alpha": 0.35,
    "g_beta": 0.2,
    "g_history": 0.25,
    "g_memory": 0.5,
    "input_q": 0.2,
    "input_history": 0.2,
    "fill_rate": 0.45,
    "tau_recovery": 5.0
  },
  "initial": {"c": 0.1, "a": 0.0, "b": 0.0, "m": 0.0, "dwell": 0.0},
  "ticks": 48
}
```


Canonical identity

Canonical JSON uses sorted keys, compact separators, ASCII escaping, and no non-finite numeric values. The model identifier is the SHA-256 digest of that serialized model. Each event digest covers its own payload, including the model identifier and previous event digest, but excludes its own digest field.

For the recorded 48-tick execution, the final event digest is

`9c90238981db6af1d474703aa5103de0`
`3a3c2c2b96776e81a159da1d7fccb696`

This identifier belongs to the emitted event chain, not to the final PDF bytes. A PDF hash containing itself is neither required nor asserted.

Reproduction contract

The reference package includes the program, input snapshot, executed outputs, and usage notes. Repeating the run in the same arithmetic environment reproduces the checked canonical record. Different Python/math-library implementations may differ in the last bits of floating-point transcendental functions and consequently in byte-level hashes; the equations and integer event conventions remain the comparison basis.